

Workflow: Building a movement track

st_as_sf() then st_transform()

records to points, in a projected CRS;
lonlat_to_utm() returns the zone

then

arrange(animal_id, timestamp)

a line is an ordered set of points, and the
order comes from the data rather than the geometry

then

st_distance(by_element = TRUE) and difftime()

the step length and the elapsed time between
consecutive records, each read against lag(), grouped
by individual so one animal is not compared with the next

then

cumsum(gap_hours > threshold)

numbers the bursts; state the threshold, and
drop the bursts holding a single fix

then

summarize(do_union = FALSE) then st_cast()

combines each burst's points in the order of
the rows, and collapses them into one line

then

st_length() then units::set_units()

measured in the units of the CRS, on the plane
for a projected CRS and the sphere for a geographic one